Module: tf.compat.v1.linalg Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/linalg

> *Public API for tf._api.v2.linalg namespace*

Documentation

Public API for tf._api.v2.linalg namespace

Modules

experimental module: Public API for tf._api.v2.linalg.experimental namespace

Classes

class LinearOperator: Base class defining a [batch of] linear operator[s].

class LinearOperatorAdjoint: LinearOperator representing the adjoint of another operator.

class LinearOperatorBlockDiag: Combines one or more LinearOperators in to a Block Diagonal matrix.

class LinearOperatorBlockLowerTriangular: Combines LinearOperators into a blockwise lower-triangular matrix.

class LinearOperatorCirculant: LinearOperator acting like a circulant matrix.

class LinearOperatorCirculant2D: LinearOperator acting like a block circulant matrix.

class LinearOperatorCirculant3D: LinearOperator acting like a nested block circulant matrix.

class LinearOperatorComposition: Composes one or more LinearOperators.

class LinearOperatorDiag: LinearOperator acting like a [batch] square diagonal matrix.

class LinearOperatorFullMatrix: LinearOperator that wraps a [batch] matrix.

class LinearOperatorHouseholder: LinearOperator acting like a [batch] of Householder transformations.

class LinearOperatorIdentity: LinearOperator acting like a [batch] square identity matrix.

class LinearOperatorInversion: LinearOperator representing the inverse of another operator.

class LinearOperatorKronecker: Kronecker product between two LinearOperators.

class LinearOperatorLowRankUpdate: Perturb a LinearOperator with a rank K update.

class LinearOperatorLowerTriangular: LinearOperator acting like a [batch] square lower triangular matrix.

class LinearOperatorPermutation: LinearOperator acting like a [batch] of permutation matrices.

class LinearOperatorScaledIdentity: LinearOperator acting like a scaled [batch] identity matrix A = c I.

class LinearOperatorToeplitz: LinearOperator acting like a [batch] of toeplitz matrices.

class LinearOperatorTridiag: LinearOperator acting like a [batch] square tridiagonal matrix.

class LinearOperatorZeros: LinearOperator acting like a [batch] zero matrix.

Functions

adjoint(...): Transposes the last two dimensions of and conjugates tensor matrix.

band_part(...): Copy a tensor setting everything outside a central band in each innermost matrix to zero.

cholesky(...): Computes the Cholesky decomposition of one or more square matrices.

cholesky_solve(...): Solves systems of linear eqns A X = RHS, given Cholesky factorizations.

cross(...): Compute the pairwise cross product.

det(...): Computes the determinant of one or more square matrices.

diag(...): Returns a batched diagonal tensor with given batched diagonal values.

diag_part(...): Returns the batched diagonal part of a batched tensor.

eigh(...): Computes the eigen decomposition of a batch of self-adjoint matrices.

eigh_tridiagonal(...): Computes the eigenvalues of a Hermitian tridiagonal matrix.

eigvalsh(...): Computes the eigenvalues of one or more self-adjoint matrices.

einsum(...): Tensor contraction over specified indices and outer product.

expm(...): Computes the matrix exponential of one or more square matrices.

eye(...): Construct an identity matrix, or a batch of matrices.

global_norm(...): Computes the global norm of multiple tensors.

inv(...): Computes the inverse of one or more square invertible matrices or their adjoints (conjugate transposes).

l2_normalize(...): Normalizes along dimension axis using an L2 norm. (deprecated arguments)

logdet(...): Computes log of the determinant of a hermitian positive definite matrix.

logm(...): Computes the matrix logarithm of one or more square matrices:

lstsq(...): Solves one or more linear least-squares problems.

lu(...): Computes the LU decomposition of one or more square matrices.

lu_matrix_inverse(...): Computes the inverse given the LU decomposition(s) of one or more matrices.

lu_reconstruct(...): The reconstruct one or more matrices from their LU decomposition(s).

lu_solve(...): Solves systems of linear eqns A X = RHS, given LU factorizations.

matmul(...): Multiplies matrix a by matrix b, producing a * b.

matrix_rank(...): Compute the matrix rank of one or more matrices.

matrix_transpose(...): Transposes last two dimensions of tensor a.

matvec(...): Multiplies matrix a by vector b, producing a * b.

norm(...): Computes the norm of vectors, matrices, and tensors. (deprecated arguments)

normalize(...): Normalizes tensor along dimension axis using specified norm.

pinv(...): Compute the Moore-Penrose pseudo-inverse of one or more matrices.

qr(...): Computes the QR decompositions of one or more matrices.

set_diag(...): Returns a batched matrix tensor with new batched diagonal values.

slogdet(...): Computes the sign and the log of the absolute value of the determinant of

solve(...): Solves systems of linear equations.

sqrtm(...): Computes the matrix square root of one or more square matrices:

svd(...): Computes the singular value decompositions of one or more matrices.

tensor_diag(...): Returns a diagonal tensor with a given diagonal values.

tensor_diag_part(...): Returns the diagonal part of the tensor.

tensordot(...): Tensor contraction of a and b along specified axes and outer product.

trace(...): Compute the trace of a tensor x.

transpose(...): Transposes last two dimensions of tensor a.

triangular_solve(...): Solve systems of linear equations with upper or lower triangular matrices.

tridiagonal_matmul(...): Multiplies tridiagonal matrix by matrix.

tridiagonal_solve(...): Solves tridiagonal systems of equations.
